Dani Roytburg

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Summary

Master's student in the Machine Learning Department at Carnegie Mellon University focused on reliable and interpretable AI, multi-agent training paradigms, and activation-based model analysis.

Education

M.S. **Carnegie Mellon University, School of Computer Science** 2027
Machine Learning

B.S. **Emory University, College of Arts and Science** 2025
Computer Science; Quantitative and Social Sciences

Positions

ML Alignment and Theory Scholars Research Fellow)
Berkeley, CA May 2026-September 2026

Advised by Shi Feng. Developed persona-training methods which emulate midtraining for better model organisms and constitutions. Engineered evaluations across 100+ training runs on synthetic-document pipelines. Evaluated how tastefully frontier models can test ideas in safety research.

Carnegie Mellon University Research Assistant, DI Group
Pittsburgh, PA September 2025-present

Develop multi-agent reinforcement learning environments to improve legibility of reasoning traces. Design user studies and adversarial setups to measure influence of legibility on human and multi-agent interactions.

Martian Research Research Fellow
San Francisco, CA May 2025-February 2026

Engineered activation profiles of language model evaluators to suppress self-preferential bias, applied to LLM routers (ICML 2026, MechInterp @ NeurIPS 2025). Winner of Mechanistic Router Interpretability Hackathon (40 submissions).

Emory University Research Assistant, Digital Humanities Lab
Atlanta, GA November 2021-May 2025

Designed and trained large-scale graph-text models; evaluated adversarial robustness of LLMs; built historical correspondence networks and dashboards.

ZS Associates Internship - Business Technology Solutions
Evanston, IL June 2024-August 2024

Automated ETL pipelines and implemented synthetic-data imputation for compromised client data.

Cloverpop Intern — Machine Learning
Chicago, IL January 2024-May 2024

Deployed decision-management systems and structured-prediction models on meeting transcripts.

McCormick and Company Intern — Global Marketing and Sales Data Science
Hunt Valley, MD June 2023-August 2023

Built Amazon sales data visualizations and automated analytics with GCP BigQuery and BERT-based embedding search.

Northeastern University Research Assistant, Mediacloud Lab
Boston, MA June 2022-October 2022

Extended NER to long-tail ethnics cuisines using Bon Appetit data.

Publications

Conference Papers

Dani Roytburg, Matthew Bozoukov, Matthew Nguyen, Jou Barzdukas, Mackenzie Puig-Hall, and Narmeen Oozeer. Are llm evaluators really narcissists? sanity checking self-preference evaluations. In *Forty-Third International Conference on Machine Learning*, 2026. URL: https://arxiv.org/abs/2601.22548

Dani Roytburg, Shreya Sridhar, and Daphne Ippolito. Measuring weak-to-strong legibility of reasoning models. In *ICML 2026 Workshop on AI for Good (AI4GOOD)*, 2026. URL: https://openreview.net/forum?id=MRStw9vqEg

Dani Roytburg and Beck Miller. Mind the gap: Pathways towards unifying ai safety and ethics research. In *Proceedings of the International Association for Safe and Ethical AI*, 2026. URL: https://arxiv.org/abs/2512.10058

Dani Roytburg, Matthew Bozoukov, Hongyu Fu, Matthew Nguyen, Jou Barzdukas, and Narmeen Fatimah Oozeer. Breaking the mirror: Activation-based mitigation of self-preference in llm evaluators. In *Mechanistic Interpretability Workshop at NeurIPS 2025*, 2025. Also accepted at: NeurIPS 2025 Workshop on Evaluating the Evolving LLM Lifecycle; NeurIPS 2025 Workshop on Reliable ML from Unreliable Data. URL: https://arxiv.org/abs/2509.03647

Dani Roytburg, Deborah Olorunisola, Sandeep Soni, and Lauren Klein. Words and action: Modeling linguistic leadership in # blacklivesmatter communities. In *Proceedings of the International AAAI Conference on Web and Social Media*, volume 19, pages 1690–1703, 2025. URL: https://ojs.aaai.org/index.php/ICWSM/article/view/35895

Thesis

Daniel Roytburg. Generative argument mining: Pretrained language models are argumentative text parsers. Undergraduate thesis, Emory University, 2025. URL: https://etd.library.emory.edu/concern/etds/5t34sm11w?locale=en

Recognition

Awards

Best Poster, LTI Student Research Symposium, 2026

Emory University Dean's List, 2022, 2024-2025

Winner, Martian Research Mechanistic Interpretability Hackathon, 2025

Skills

Python Java R JavaScript Typescript SQL PyTorch JAX HuggingFace Transformers scikit-learn spaCy networkx D3.js React.js GCP Docker MySQL